

A cubic obstruction to Conjecture 00000003315

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Abstract

The toric ideal of the full lattice-point configuration of the triangle $\text{conv}\{(0,0), (2,1), (1,2)\}$ contains the nonzero cubic $X_1X_2X_3 - X_0^3$, but contains no nonzero binomial of degree at most two. It therefore cannot be generated by quadratic circuit relations, as asserted in Conjecture 00000003315. The accompanying Lean project proves this for an actual polynomial algebra map, its kernel ideal, and every putative set of quadratic binomial generators.

1 The lattice polygon and its toric map

Let

$$P = \text{conv}\{(0,0), (2,1), (1,2)\}.$$

Its complete set of lattice points, in the order used below, is

$$p_0 = (1,1), \quad p_1 = (2,1), \quad p_2 = (1,2), \quad p_3 = (0,0).$$

Indeed, a point of P has the form $(x,y) = (2b+c, b+2c)$ with $b, c \geq 0$ and $b+c \leq 1$. Thus

$$x, y \geq 0, \quad x+y \leq 3, \quad x \leq 2y, \quad y \leq 2x.$$

For integral x, y , either one coordinate is zero and both are zero, or both are positive and $x+y \leq 3$ gives exactly $(1,1), (2,1), (1,2)$. Conversely the three vertices belong to P , and $(1,1)$ is their barycenter. This proves the claimed complete lattice-point list.

Homogenize the points to the columns $a_i = (1, p_i)$. Over $\mathbb{Q}$, consider the polynomial algebra homomorphism

$$\begin{aligned} \varphi : \mathbb{Q}[X_0, X_1, X_2, X_3] &\longrightarrow \mathbb{Q}[t, x, y], \\ (X_0, X_1, X_2, X_3) &\longmapsto (txy, tx^2y, txy^2, t). \end{aligned}$$

The toric ideal of this homogenized polytopal lattice configuration is $I = \ker \varphi$. The homogeneous first coordinate is essential here: in particular, $p_3 = (0,0)$ maps to t , not to the constant 1.

Proposition 1. *I is nonzero and has no generating set consisting of binomials of degree at most two. In particular, it is not generated by quadratic circuit relations.*

Proof. The distinct monomials in

$$F = X_1X_2X_3 - X_0^3$$

have the same image $t^3x^3y^3$. Hence $0 \neq F \in I$.

For $e = (e_0, e_1, e_2, e_3) \in \mathbb{N}^4$, the image of X^e is the monomial with exponent vector

$$Ae = (e_0 + e_1 + e_2 + e_3, e_0 + 2e_1 + e_2, e_0 + e_1 + 2e_2).$$

Suppose $Ae = Af$ and $|e|, |f| \leq 2$. Writing $d = e - f \in \mathbb{Z}^4$, the three coordinate equations give

$$d = (-3r, r, r, r) \quad \text{for some } r \in \mathbb{Z}.$$

If $r > 0$, then $f_0 = e_0 + 3r \geq 3$, contrary to $|f| \leq 2$. If $r < 0$, then $e_0 = f_0 - 3r \geq 3$, contrary to $|e| \leq 2$. Thus $r = 0$ and $e = f$.

Now let $G = \alpha X^e - \beta X^f \in I$ with $|e|, |f| \leq 2$ and $\alpha, \beta \in \mathbb{Q}$. In the target polynomial ring, $\alpha t^{(Ae)_0} x^{(Ae)_1} y^{(Ae)_2} = \beta t^{(Af)_0} x^{(Af)_1} y^{(Af)_2}$. Equality of monomials implies either both coefficients vanish, or $Ae = Af$ and $\alpha = \beta$. In either case $G = 0$, by the injectivity just proved.

If a set S of such binomials generated I , each member of S would lie in I and hence be zero. Its generated ideal would then be zero, contradicting $F \in I \setminus \{0\}$. Quadratic circuit relations are a particular class of quadratic binomials in I , so they cannot generate I either. $\square$

2 Scope and formal verification

The source [1] asserts quadratic circuit generation in both languages. Our configuration includes *all* lattice points of an explicit lattice polytope. The failure of this generation claim disproves the conjecture; no conclusion about its Markov-basis clause is needed.

In `lean/Main.lean`, `Source` and `Target` are actual Mathlib multivariate polynomial rings over $\mathbb{Q}$. `toricMap` is an algebra homomorphism defined by polynomial evaluation, and `toricIdeal` is its actual `RingHom.ker`. The formal proof includes:

- `configuration_is_exact_triangle` proves that the columns come from exactly all the integer points in the triangle, using real nonnegative barycentric coordinates.
- `toricMap_monomial` proves substitution for arbitrary exponents and coefficients. The coordinate sum is proved equal to the ordinary exponent sum.
- The cubic is proved equal to $X_1 X_2 X_3 - X_0^3$, nonzero, and in the kernel. Every kernel binomial whose two terms have degree at most two is proved zero.
- `no_quadratic_binomial_generating_set` quantifies over every set of such polynomials and rules out equality of its `Ideal.span` with the kernel. No classification of arbitrary quadratic polynomials or Markov bases is claimed.

The pinned project uses Lean 4.19.0 and Mathlib. Run in `lean/`:

```
lake build
lake env lean -DwarningAsError=true Main.lean
```

The supplementary `verify.py` checks the lattice points, all fifteen monomials of degree at most two, and the cubic with exact arithmetic. Lean proves the universal ideal statement. Build logs, axiom audits, and hashes are in `verification/`; the exact original source bytes are in `SOURCE.md`.

References

- [1] The Last Math Competition, conjecture 00000003315, repository statement.